

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		all three needed chloride – white bromide – cream (accept off-white) yellow – iodide	1			1		1
		(ii)		$\Delta_f H^\theta (\text{AgBr}) = 105.6 + (\frac{1}{2} \times -243.1) - 84.4$ (1) $\Delta_f H^\theta (\text{AgBr}) = -100.35$ (1) AgCl more stable than AgBr (ecf possible from values) (1) as standard enthalpy of formation is more negative (1)		2	2	4	1	
	(b)	(i)	I	$\text{Cu} + 2\text{Ag}^+ \rightarrow \text{Cu}^{2+} + 2\text{Ag}$		1		1		
			II	$E^\theta (\text{Cu})$ must be more negative/less positive than +0.80V (1) award (1) for either of following reasons reaction is feasible when EMF is positive reaction is feasible when reduction half-cell is more positive than oxidation half-cell			2	2		

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		(ii)	I	award (2) for correct answer $E^{\ominus} = 0.80 - 0.93 = -0.13 \text{ V}$ if incorrect award (1) for use of 0.93 V		1	1	2	1	
			II	as concentration of Ag^+ decreases the $\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag}$ equilibrium shifts to the left (1) this makes E for this reaction less positive (1) EMF of the cell is $E_{\text{Ag}} - E_{\text{Pb}}$ so the value becomes less positive (1)			3	3		
				Question 9 total	1	4	8	13	2	1